Linux File Types in `ls -l`

When using `ls -l`, the first character in the output indicates the **file type**. Below are all possible symbols and their meanings, with examples.

File Type Symbols

Symbol	Type	Description
`-`	Regular file	Normal files (text, binary, images, executables, etc.)
`d`	Directory	Folder containing files and other directories
`l`	Symbolic link	A shortcut or reference to another file or directory
`c`	Character device file	Device file that provides serial I/O (e.g., keyboard, mouse, `/dev/tty`)
`b`	Block device file	Device file for block storage (e.g., hard drives `/dev/sda`)
`p`	Named pipe (FIFO)	Special file for inter-process communication
`s`	Socket	Used for communication between processes (local or over network)

Examples of Each File Type

1. Regular File (`-`)

```
ls -l
-rw-r--r--  1 user user  1234 Sep 18 10:00 file.txt
```

Explanation: - `-` → Regular file - Example: `file.txt` is a normal text file.

Regular files can be: - Text files (e.g., `.txt`, `.md`) - Binary files (e.g., `.exe`, `.bin`) - Media files (e.g., `.png`, `.mp3`)

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2. Directory (`d`)

```
ls -l
 drwxr-xr-x  2 user user  4096 Sep 18 09:58 Documents/
```

Explanation: - `d` → Directory (folder) - Example: `Documents/` is a folder containing files or subdirectories.

3. Symbolic Link (`l`)

```
ls -l
lrwxrwxrwx  1 user user   11 Sep 18 09:59 shortcut -> file.txt
```

Explanation: - `l` → Symbolic link (symlink) - Example: `shortcut` points to `file.txt` - Acts like a shortcut, if the target file is deleted, the link becomes broken.

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4. Character Device File (`c`)

```
ls -l
crw-rw----  1 root tty   4, 64 Sep 18 09:55 /dev/ttyS0
```

Explanation: - `c` → Character device - Provides serial I/O, sending data one character at a time. - Example: `/dev/ttyS0` (serial port, terminal, mouse, keyboard).

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5. Block Device File (`b`)

```
ls -l
brw-rw----  1 root disk  8, 0 Sep 18 09:55 /dev/sda
```

Explanation: - `b` → Block device - Handles data in blocks, typically for storage devices. - Example: `/dev/sda` (hard disk drive, SSD, USB drive).

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6. Named Pipe (FIFO) (`p`)

```
ls -l
prw-------  1 user user     0 Sep 18 10:01 mypipe
```

Explanation: - `p` → Named pipe (FIFO) - Used for **inter-process communication (IPC)** - Example: `mypipe` can transfer data between two running processes.

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7. Socket (`s`)

```
ls -l
srwxr-xr-x  1 user user     0 Sep 18 10:02 my_socket
```

Explanation: - `s` → Socket file - Enables communication between processes (local or over the network). - Example: `my_socket` can be used by applications like databases, web servers.

Quick Reference Summary

- `-` → Regular file
- `d` → Directory
- `l` → Symbolic link
- `c` → Character device (keyboard, mouse, terminals)
- `b` → Block device (hard drives, SSDs)
- `p` → Named pipe (FIFO, IPC)
- `s` → Socket (process communication)

✅With `ls -l`, always check the **first character** to identify the file type.